SMART AUTONOMOUS IN-BASKET RECOGNITION SHOPPING CART

An autonomous cart for intelligent product recognition and automated checkout.

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FROM MANUAL SHOPPING TO AUTONOMOUS CHECKOUT

- Traditional shopping requires manual cart pushing and can involve long checkout queues.
- Manual product scanning can introduce errors and slow the shopping experience.
- The proposed cart navigates supermarket aisles independently while avoiding obstacles.
- IoT, load-cell sensor fusion, and an internal camera enable instant automated billing.



LITERATURE REVIEW

Shanmugapriya et al. (2021)

Contribution

RFID and IoT smart cart focused on automated billing.

Technology

Ali et al. (2019)

Contribution

Raspberry Pi-based cart architecture for item-data processing.

Technology

Al-Kassab et al. (2020)

Contribution

automated Point of Sale system integration.

Lu et al. (2023)

Contribution

indoor mobile-robot navigation using ROS and sensor fusion.

Technology



COMPARISON OF STUDIED PAPERS

Paper	Primary Technology	Key Advantage	Limitation
Shanmugapriya et al. (2021)	RFID & IoT	Fast item scanning	Requires manual line-of-sight scanning
Ali et al. (2019)	Raspberry Pi	Embedded processing unit	Lacks indoor obstacle avoidance
Al-Kassab et al. (2020)	Smart POS System	Automated billing integration	Manual cart handling required
Lu et al. (2023)	ROS & Sensor Fusion	High navigation accuracy	Does not include checkout mechanisms
Proposed System	In-Basket Vision + Load Cell + ROS	Autonomous navigation & dual fraud verification	Requires initial AI training datasets



PROJECT TIMELINE & DEVELOPMENT PHASES



Phase 1

Current Stage:

Extracting and analyzing research paper ideas, evaluating their pros and cons, and identifying potential technical challenges.



Phase2

Development Stage:

Implementing the core project idea featuring the in-basket camera system for automated product recognition and autonomous navigation.



Phase 3

Simulation & Finalization:

Conducting comprehensive simulation testing of the project to validate performance and functionality.

PROJECT OBJECTIVES

Design an autonomous, self-navigating shopping cart with an integrated AI visual recognition system for automated checkout.



Obstacle avoidance

Operate in dynamic store environments.

Dual verification

Combine a weight sensor and in-basket AI camera to help prevent fraud.

Real-time billing

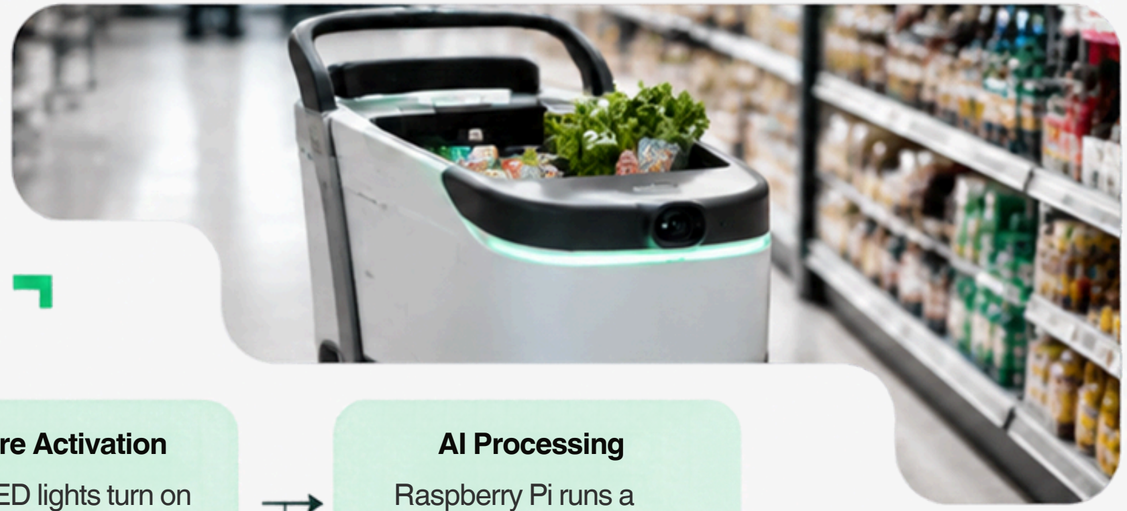
Display live totals through an interactive user touchscreen.

TECHNICAL SPECIFICATIONS

Component / Subsystem	Specification / Details
Main Processor	Raspberry Pi (executes AI models and UI)
Visual Sensor	High-definition internal in-basket camera
Weight Sensor	Load Cell with HX711 Amplifier Module
Navigation System	Sensor Fusion and ROS (Robot Operating System)
User Interface	On-board touchscreen display



PROJECT EXECUTION FLOWCHART



Start / Idle State

cart navigates store aisles while avoiding static and dynamic obstacles.

Item Trigger

customer places an item in the basket; Load Cell (HX711) detects a weight shift.

Hardware Activation

internal LED lights turn on and the internal camera captures the item.

AI Processing

Raspberry Pi runs a computer vision model to identify the product.



Dual Verification

does measured weight match the AI-identified product's database weight?

No

Prompt customer to rescan item

Yes

Update running total on touchscreen

Checkout Completion

final transaction processed automatically.

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An IoT-Based Smart Shopping Cart Using the Contemporary Barcode Scanner

V. N. Prithvish, Shraddha Agrawal and John Sahaya Rani Alex

Abstract An efficacious prototyping of an intelligent Internet of Things (IoT)-based smart cart that primarily enhances the shopping experience of the customers as well as owners. The assembly of the smart cart consists of a contemporary barcode scanner from the present-day shopping scenario, feedback systems providing product weight and product imaging to avoid discrepancies, a cloud-based database and an embedded hardware to connect the above. The mechanism facilitates the customer to add and remove products throughout the shopping, updating the bill instantaneously, and further reflecting changes in the inventory. The framework incorporates a Parse cloud-based inventory to ease the managerial task at the owner end. Further focus has been also given on the proper placement of all the components on a cart for the efficient working of the shop, keeping user's ease of usage in mind.

Keywords Smart cart · IoT · Pars · Beagle Bone Black · TVS electronics BSC-101 star · RTL-8188EU · SIFT algorithm · Load cell · Barcode scanner

1 Introduction

Retail market is growing at an alarming rate, and the modern customer needs better than the best experience. Technology will play a very important role in improving and developing the experience which sets the modern retail apart from the brick and

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Smart Shopping Trolley with Auto Billing System Using RFID

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Keywords: RFID, Smart Trolley, Auto Billing, Retail Automation, IoT.

Abstract: This study investigates employing RFID-enabled intelligent shopping trolleys within modern retail environments, paying special attention to nascent markets. Using an integrated framework that merges technology acceptance models with metrics for operational efficiency, the analysis assesses user adoption patterns relative to system performance. We present controlled experiment data from 200 subjects in a variety of retail environments that identifies important factors for technology adoption (e.g., use, ease-of-use, and system reliability). We show that RFID-based automated billing can indeed reduce checkout time significantly (by about 70%) but its implementation in the real world presents several challenges in terms of infrastructure requirements and initial investment costs. The study shows consumer preference for automated self-checkout solutions consistently increases over time, with the rate of this increase tempered by factors like user technology familiarity and retailer willingness. These explorations also help us better understand technological transformation in retail ecosystems, notably in developing markets where such implementations are still new. Incorporating practical constraints that could hinder smart shopping systems' implementation are thoroughly examined, along with many suggestions for improvement.

1 INTRODUCTION

Technology has advanced over the past few years changing industries, including retail. Increasingly, innovative systems have been developed to remove the inefficiencies of traditional shopping methods, driven by the demand for more efficient, easy-to-use, and automated techniques. One such inefficiency is the traditional billing system, where the customers manually select the products and waiting in long lines at checkout counters for considerable time, and may be subjected to billing errors as a result of human intervention. It is a lengthy and labour intensive process which takes away from the consumer shopping experience. RFID (Radio Frequency Identification) Using Smart Shopping Trolley with Auto Billing System was introduced to overcome these shortcomings. The solution aims to automate the complete shopping and checkout experience to reduce human intervention and streamline operations. Unlike traditional barcodes, which need to be scanned manually, RFID tags are read at a distance and processed in real-time. This allows the system to automatically sense items entering or leaving the trolley. The in-real-time digital depictions of the total bill price as products are being scanned, to continuously show customers their current cost is one of the relevant features of the proposed system presented in this topic. It also gives customers greater control over their purchases with transparency and control as they check out. The contactless self-checkout option also does away with queuing: shoppers simply head to payment after their purchases. Seamlessly integrated with modern payment systems, it saves time and makes shopping easier. The system provides additional benefits to retailers apart from improving the customer experience. Automated billing also ensures better accuracy in transactional records, reducing efforts for discrepancies and manual work. Aside from this, item detection, and transaction tracking can also help, which gives real-time data about stock at hand and better business decisions. The introduction of intelligent systems into retailing is a step toward automation and digitalization, enabled by the Internet of Things (IoT) and RFID technologies. Innovative solutions are fairly easy to implement in life, so this intelligent shopping trolley is a game changer that

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FULL LENGTH ARTICLE

A Smart Retail Point-of-Sale System Using Machine Learning Forecasting and Application-Level Failover

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ABSTRACT

Point-of-Sale (POS) systems support retail transaction processing, inventory management, and operational decision-making. However, many traditional POS systems rely on a single centralized database, creating a critical point of failure during server faults, database errors, or network disruptions. Conventional POS systems may also lack intelligent forecasting capabilities, causing inventory decisions to depend on manual estimation or fixed reorder rules. These limitations can lead to operational interruption, inventory inconsistency, and inefficient stock management. This study proposes a high-availability smart POS system integrated with AI-based inventory forecasting. The framework uses a dual-database architecture consisting of a primary database and a secondary backup database. Application-level failover redirects operations automatically when the primary database becomes unavailable, while an event-driven backup mechanism synchronizes important data after critical transactions. The forecasting component was developed using the Rossmann Store Sales dataset. Several models were evaluated, including Random Forest (RF), Bagging, CatBoost, XGBoost, and KNN Regressor. Optimization algorithms including PSO, MVO, HHO, and DE were also integrated with RF. The results showed that baseline RF achieved the best overall performance, with MSE = 0.080454, RMSE = 0.283643, MAE = 0.184264, R^2 = 0.919712, and WI = 0.978542. Among optimized models, PSO + RF achieved the strongest optimizer-based result, with MSE = 0.081480 and RMSE = 0.285446. The implemented system also demonstrated successful database connectivity, failover readiness, and automatic backup execution. The findings confirm that the proposed framework can support continuous retail operation while providing accurate AI-based forecasting and inventory recommendations.

Keywords: Smart POS • Machine learning forecasting • Application-level failover • Inventory management • Random Forest

1. INTRODUCTION

Point-of-Sale (POS) systems have become fundamental components of modern retail environments because they support transaction processing, inventory management, sales monitoring, and operational decision-making. Contemporary POS platforms are no longer limited to billing and payment activities; rather, they have evolved into intelligent business management systems that integrate transactional functions with analytical and predictive capabilities [1, 2]. The digital transformation of retail has increased organizational dependence on POS infrastructures for operational continuity, customer experience improvement, and real-time business intelligence [3, 4]. Consequently, reliability, availability, and forecasting

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Review

Sensor-Fusion Based Navigation for Autonomous Mobile Robot

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Abstract: Navigation systems are developing rapidly; nevertheless, tasks are becoming more complex, significantly increasing the number of challenges for robotic systems. Navigation can be separated into global and local navigation. While global navigation works according to predefined data about the environment, local navigation uses sensory data to dynamically react and adjust the trajectory. Tasks are becoming more complex with the addition of dynamic obstacles, multiple robots, or, in some cases, inspection of places that are not physically reachable by humans. Cognitive tasks require not only detecting an object but also evaluating it without direct recognition. For this purpose, sensor fusion methods are employed. However, sensors of different physical nature sometimes cannot directly extract required information. As a result, AI methods are becoming increasingly popular for evaluating acquired information and for controlling and generating robot trajectories. In this work, a review of sensors for mobile robot localization is presented by comparing them and listing advantages and disadvantages of their combinations. Also, integration with path-planning methods is looked into. Moreover, sensor fusion methods are analyzed and evaluated. Furthermore, a concept for channel robot navigation, designed based on the research literature, is presented. Lastly, discussion and conclusions are drawn.

Keywords: sensor fusion; mobile robot; navigation; machine learning

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1. Introduction

In the rapid development of the automation and robotics world, making of autonomous vehicles and mobile robots is a big step toward operational efficiency, safety, and autonomy. At the heart of this technological revolution is the intricate domain of sensor fusion, a paradigm that merges data from different sensors to make cohesive and accurate perceptions of operational environments. This paper goes into the realm of sensor-fusion-based navigation systems for autonomous robots, spotlighting diverse methodologies that underpin their functionality and emerging trends that shape their evolution.

Navigational autonomy in robots is paramount for their effective deployment across a spectrum of applications, from industrial automation to exploration in inaccessible terrains. Traditional navigation methodologies, while foundational, often grapple with complexities and dynamic changes intrinsic to real-world environments. Bridging this gap, advanced navigation systems harness the synergy of global and local navigation methods [1]. Global navigation operates on the premise of pre-acquired environmental knowledge, facilitating formulation and adherence to predetermined paths. In contrast, local navigation equips mobile robots with agility to dynamically refine their paths in real time, utilizing an arsenal of external sensors—ranging from infrared and ultrasonic sensors to LASER, LIDAR, and cameras [2]. This sensorial diversity, when orchestrated by sophisticated software